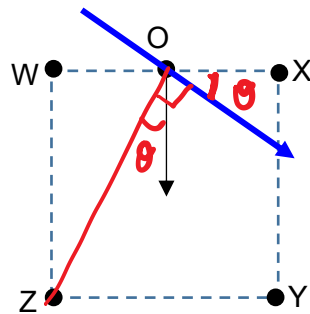


a resultant force.

- 23 Four parallel current carrying conductors are placed vertically at the corners of a square WXYZ. The magnitudes of the current at W and X are I while the magnitudes of the current at Y and Z are $2I$.



$$\sin \theta = \frac{d/2}{\sqrt{5}d/2} = \frac{1}{\sqrt{5}}$$

O is the mid-point between W and X.

If there is to be a resultant magnetic flux density produced at O in the direction shown, which option gives the largest magnetic flux density?

	current into the plane	current out of the plane
A	W and Z	X and Y
B	X and Z	W and Y
C	W and Y	X and Z
D	X and Y	W and Z

Ans: A

Let the side of square be d.

$$\text{B-field due to W} = \mu_0 I / (2\pi d/2) = (2 \mu_0 I) / (2\pi d)$$

$$\text{B-field due to X} = \mu_0 I / (2\pi d/2) = (2 \mu_0 I) / (2\pi d)$$

$$\text{B-field due to Z} = \mu_0 2I / (2\pi \sqrt{5} d/2) = (4/\sqrt{5} \mu_0 I) / (2\pi d)$$

$$\text{B-field due to Y} = \mu_0 2I / (2\pi \sqrt{5} d/2) = (4/\sqrt{5} \mu_0 I) / (2\pi d)$$

For Z, the B field will be tangent to the point O when you draw a circle about Z cutting O. To find the component perpendicular to WX,

$$\text{B-field component due to Z} = (4/\sqrt{5} \mu_0 I) / (2\pi d) \sin\theta = (4/5 \mu_0 I) / (2\pi d).$$

$$\text{Likewise for Y, B-field component due to Z} = (4/5 \mu_0 I) / (2\pi d)$$

The largest magnetic flux density will be W and Z into the plane and X and Y out of the plane.